Excess entropy and divalent-ion mobility in multicomponent charged LJ mixtures

HNC screening · replicated MD · thermodynamic integration, pair entropy and 2PT

+0.30 kB

excess entropy gained by HNC-selected solvent designs

+18–20%

divalent-ion self-diffusion, both signs

0%

effect of adding labels without changing interactions

Design: label count vs interaction diversity

1

HNC screen

35 integer-charge, composition and dispersion candidates at $T^* = 2$, $\rho^* = 0.7$; 8 selected

2

Replicated MD

8 mixtures × 3 matched replicas, N = 4000, x(±2) = 0.03; probes matched across mixtures

3

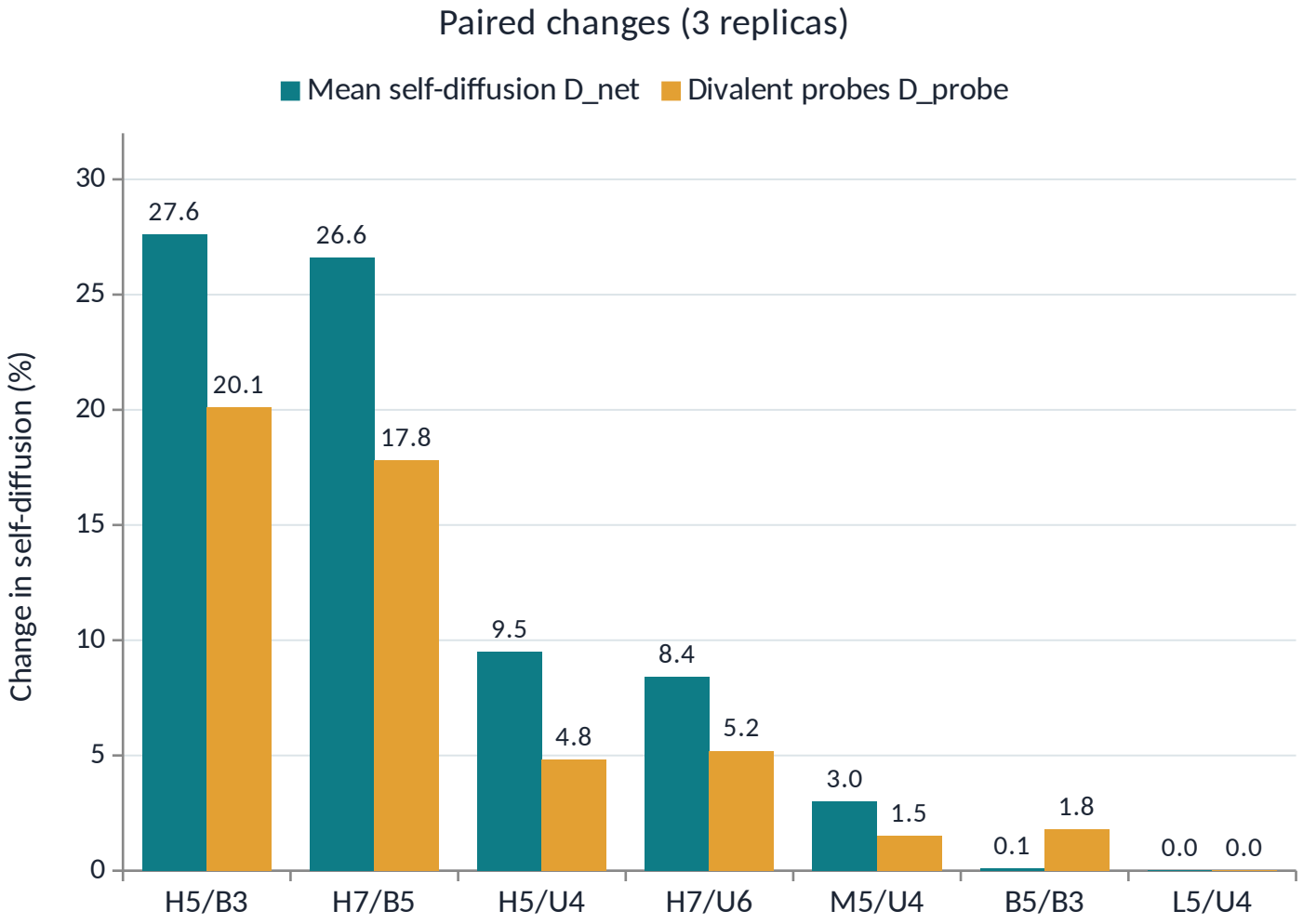
Entropy and transport

TI over 19 densities (fixed LJ cutoff, Debye–Hückel endpoint); MSD in one common COM frame

Code	Mixture (all with 3% of each ±2 probe)	Role
B3	94% LJ solvent	Baseline
B5	B3 + 5% of each ±1 ion	Salt addition
U4	64% solvent + 30% weak solvent, ε = 0.25	Reference
L5	U4 with 2 identical weak-solvent labels	Label null
M5	U4 split into ε = 0.0625 / 0.5625	Moderate split
H5	U4 split into ε = 0.01 / 0.81	Strong split
U6 / H7	U4 / H5 plus 5% of each ±1 ion	With salt

U4 → M5 → H5 keep charged composition, ionic strength and the mean bare LJ amplitude $\Sigma_{ij} x_i x_j \varepsilon_{ij} = 0.7225$ fixed

Entropy speeds up ions, less than the solvent



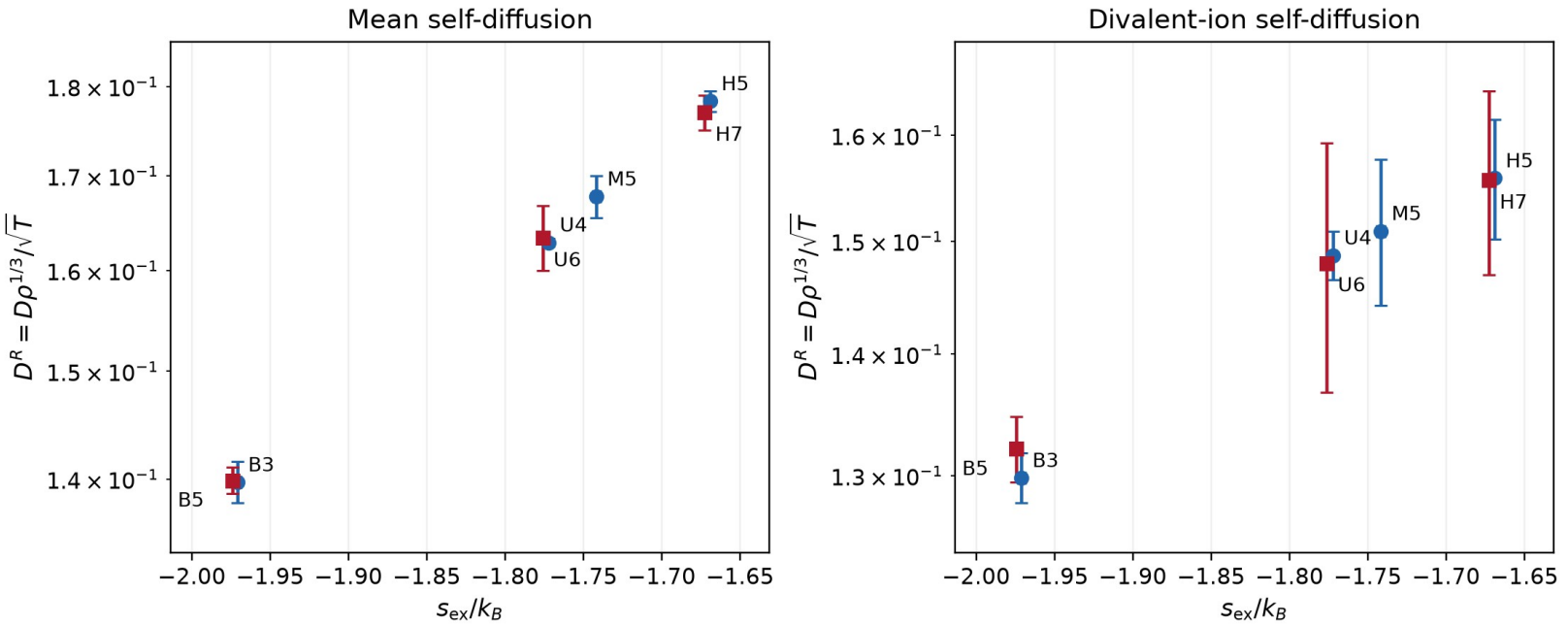
Full designs: Δs_ex ≈ +0.30 kB, probes +18–20%; each sign accelerates separately

Controlled diversity: Δs_ex = +0.103 kB, mean +8–9%, probes ≈ +5% (window-sensitive)

Labels only (L5/U4): identical diffusion in every replica

Salt only (B5/B3): Δs_ex slightly negative, no resolved speedup

TI entropy tracks mobility; HNC and s2 follow



Descriptive Rosenfeld fits (3–4 mixtures, $R^2 > 0.99$): B_net ≈ 0.8, B_probe ≈ 0.6. Controlled local probe slope ≈ 0.5, about half the mean-fluid value. Absolute HNC entropies are 0.06–0.09 kB too low; increments are much better.

≤ 7%

HNC error on the controlled entropy increments

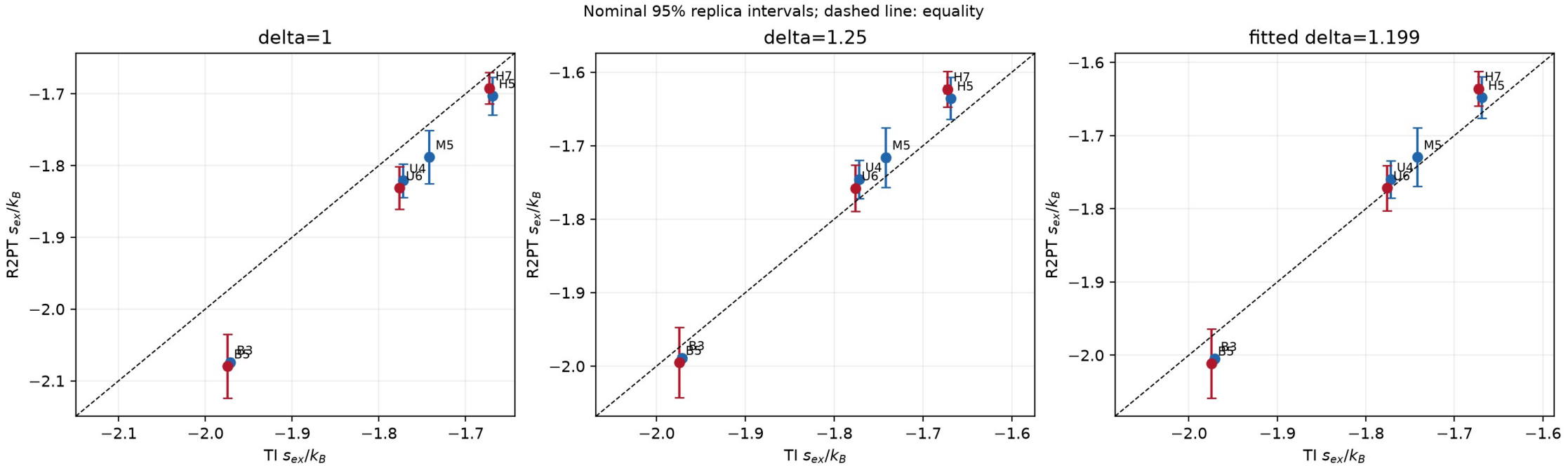
≈ 80%

share of s_ex carried by the MD pair entropy s_2, and of its increments

−0.33 kB

beyond-pair residual s_ex − s_2: composition dependent, not negligible

2PT when TI is unfeasible: calibrate, then rank



Calibrate one δ

Fit over 7 mixtures: $\delta^* = 1.20$ [1.17, 1.23]; $\delta = 1.25$ cuts RMS bias 0.067 → 0.029 kB; held-out bias 0.030 kB

Good within a family

M5/U4 and H5/U4 increments reproduced within 0.01 kB; label split exactly zero

Limits

H7/U6 +30%, baseline → design +17–23%; D enters the fluidicity, so 2PT cannot test an entropy–diffusion link on its own

Next: fast entropy estimators for electrolytes

2PT, velocity based

Ready

- Calibrated δ ≈ 1.2–1.25 for this model family
- One dense trajectory per candidate
- Use for ranking; recalibrate on TI anchors
- Not independent of diffusion

Voronoi cluster entropy

Open

- 24 trajectories packaged with TI, s_2 and D references
- Must reproduce TI order, the 0.03 and 0.10 kB increments and the label null
- Structure based, so independent of transport

1. **HNC screen**
   candidate mixtures
2. **TI anchors**
   2–3 states to calibrate and validate
3. **2PT / Voronoi**
   rank many candidates
4. **Conductivity**
   Green–Kubo current, not only self-diffusion

In progress: 6 longer replicas (5000 τ) and N = 8000 runs of B3, U4, H5, U6, H7 to settle the ≈5% controlled ion effect